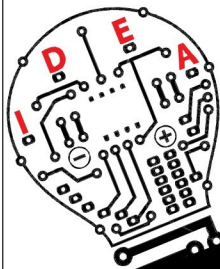


electronics
FOR YOU

Original, innovative circuit ideas are invited

from designers and
experimenters
for publication in EFY magazine



**Honorarium (minimum ₹1,000)
will be paid to the contributors soon
after publication of their articles.**

Complete details may be sent to the
Editor, Electronics For You,
D-87/1, Okhla Industrial Area Phase 1,
New Delhi 110020
or emailed to editsec@efy.in

For editorial guidelines, visit
http://efymag.com/Guidelines_for_EFY_authors.doc

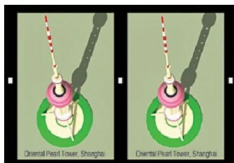


Fig. 2: A set of stereoscopic images

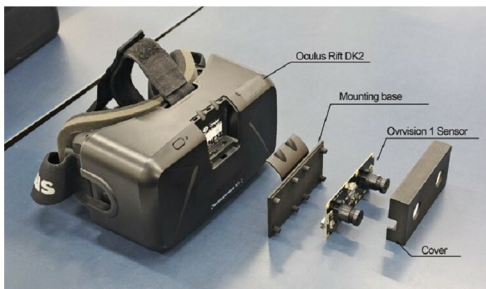


Fig. 3: Oculus Rift setup

with two different colour bases (cyan and red). This essentially explains the science behind the working of a stereo camera.

Constructing an image

Based on the above principle, a stereo camera is designed to perceive depth using two cameras. These cameras are separated by a distance called baseline, which is nothing but the interocular distance. Usually this distance varies anywhere between 55mm and 85mm.

A camera is nothing but an image sensor, which actively converts light into electrical signals. This is basically analogue-to-digital conversion, and a single frame of image that is captured is converted into a stream of bits.

There are many ways in which these bits are delivered from the image sensor. A few examples are parallel interface, low-voltage differential signalling (LVDS), MIPI interface and HiSpi interface.

In a stereo camera, there are two streams of data bits originating from two cameras. The complexity arises here. There are two image sensors focusing on a single object and the expectation is that, these two image sensors produce exactly the same output, meaning, the same stream of bits. But since the two cameras are separated by some distance, even

though both cameras focus on the same object, there is a slight difference in both sets of outputs. Fig. 2 shows such a set of images captured using a stereo camera.

These two different streams of output are then fed to the processor, which accepts stereoscopic inputs. Note that, not all processors available in the market support stereo input. The processor treats this difference between the two images and, using depth-perceiving algorithms, calculates the depth or the distance at which that particular object is placed. This is the simple working principle of a stereo camera, which is used in a system like Oculus Rift, whose complete setup is shown in Fig. 3.

While stereo cameras are a great source for 3D images or videos, these are not the only source using which 3D recreation is done for AR. Another interesting way of recreating 3D images is using a concept called time of flight (ToF). ●